E Ink 元太科技 E Ink Holdings

PRODUCT INFORMATION

6.25 ” flexible EPD

TENTATIVE

Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	6.25	Inch	
Display Resolution	200 (H)×1040(V)	Pixel	
Active Area	30.0 (H)×156.0 (V)	mm	
Pixel Pitch	0.15 (H) × 0.15 (V)	mm	
Pixel Configuration	Rectangle		
Outline Dimension	40.3 (W) × 173.2 (H) ×1.89 (D)	mm	
Module Weight	1.7	g	
Number of Gray	16 Gray Level (monochrome)		
Display operating mode	Reflective mode		

Absolute maximum rating

Parameter	Symbol	Rating	Unit	Remark
Logic Supply Voltage	VDD	-0.3 to +5	V	--
Positive Supply Voltage	V_{POS}	-0.3 to +18	V	--
Negative Supply Voltage	V_{NEG}	+0.3 to -18	V	--
Max .Drive Voltage Range	V_{POS} - V_{NEG}	36	V	--
Supply Voltage	VGG	-0.3 to +46	V	--
Supply Voltage	VGL	-25.0 to +0.3	V	--
Supply Range	VGG-VGL	-0.3 to +46	V	--
Operating Temp. Range	TOTR	TBD	℃	--
Storage Temperature	TSTG	TBD	℃	--

Panel DC characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Signal ground	V_{SS}		-	0	-	V
Logic Voltage supply	V_{DD}		3.0	3.3	3.6	V
	I_{VDD}	V_{DD}=3.3V	-	TBD	TBD	mA
Gate Negative supply	V_{GL}		-21	-20	-19	V
	I_{GL}	V_{GL} =-20V	-	TBD	TBD	mA
Gate Positive supply	V_{GG}		21	22	23	V
	I_{GG}	V_{GG} = 22V	-	TBD	TBD	mA
Source Negative supply	V_{NEG}		-15.4	-15	-14.6	V
	I_{NEG}	V_{NEG} = -15V	-	TBD	TBD	mA
Source Positive supply	V_{POS}		14.6	15	15.4	V
	I_{POS}	V_{POS} = 15V	-	TBD	TBD	mA
Border supply	V_{COM}		TBD	Adjusted	TBD	V
Asymmetry source	V_{Asym}	V_{POS}+V_{NEG}	-800	0	800	mV
Common voltage	V_{COM}		TBD	Adjusted	TBD	V
	I_{COM}		-	TBD	-	mA
Panel Power	P		-	TBD	TBD	mW
Standby power panel	P_{STBY}		-	-	TBD	mW

Display Module AC characteristics

VDD=3.0V to 3.6V, unless otherwise specified.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Clock frequency	fckv	-	-	200	kHz
Minimum “L” clock pulse width	twL	0.5	-	-	us
Minimum “H” clock pulse width	twH	0.5	-	-	us
Clock rise time	trckv	-	-	100	ns
Clock fall time	tfckv	-	-	100	ns
SPV setup time	tSU	100	-	twH-100	ns
SPV hold time	tH	100	-	twH-100	ns
Pulse rise time	trspv	-	-	100	ns
Pulse fall time	tfspv	-	-	100	ns
Clock XCL cycle time	tcy	16.67	-	-	ns
D0 .. D7 setup time	tsu	8	-	-	ns
D0 .. D7 hold time	th	8	-	-	ns
XSTL setup time	tstls	0.5* tcy	-	0.8* tcy	ns
XSTL hold time	tstlh	0.5* tcy	-	160*tcy-tstls	ns
XLE on delay time	tLEdly	3.5* tcy	-	-	ns
XLE high-level pulse width (When VDD=3.0V to 3.6V)	tLEw	300	-	-	ns
XLE off delay time	tLEoff	200	-	-	ns
Output setting time to +/- 30mV(C_{load}=200pF)	tout	-	-	20	us

Connector type: PANASONIC AXE534124

Pin assignment

Pin	Symbol	Description of pins
1	VNEG	Negative power supply source driver
2	VPOS	Positive power supply source driver
3	VNEG	Negative power supply source driver
4	VPOS	Positive power supply source driver
5	NC	No connection
6	NC	No connection
7	VDD	Digital power supply drivers
8	VDD	Digital power supply drivers
9	VDD	Digital power supply drivers
10	VSS	Ground
11	NC	No connection
12	XOE	Output enable source driver
13	XCL	Clock source driver
14	XSTL	Start pulse source driver
15	XLE	Latch enable source driver
16	VSS	Ground
17	D0	Data signal source driver
18	D1	Data signal source driver
19	D2	Data signal source driver
20	D3	Data signal source driver
21	D4	Data signal source driver
22	D5	Data signal source driver
23	D6	Data signal source driver
24	D7	Data signal source driver
25	CKV	Clock gate driver
26	SPV	Start pulse gate driver
27	MODE1	Output mode selection gate driver
28	VSS	Ground
29	VCOM	Common connection
30	BORDER	Border connection
31	NC	No connection
32	NC	No connection
33	VGH	Positive power supply gate driver
34	VGL	Negative power supply gate driver

Mechanical Drawing of EPD Module